

1) Use a decision structure to write an appropriate statement for each of the following:

a) Display “Great Job!” when grade is 90 or higher

```
if (grade >= 90) {  
    System.out.println(“Great Job!”);  
}
```

b) Display “error” when number is less than 20 or greater than 50

```
if (number < 20 || number > 50) {  
    System.out.println(“error”);  
}
```

c) Add 2 to the value of y when y is less than 100

```
if (y < 100) {  
    y += 2;  
}
```

2) Assume num1 and num2 contain integer values. Write an if-else statement that displays one of the following messages as appropriate:

First number is larger.

Second number is larger.

Numbers are equal.

```
if (num1 > num2) {  
    System.out.println(“First number is larger.”);  
} else if (num2 > num1) {  
    System.out. println(“Second number is larger.”);  
} else {  
    System.out. println(“Numbers are equal.”);  
}
```

3)

a) Which is the appropriate word, odd or even for the blanks below?

```
If (num % 2 == 0) {  
    System.out.println(“even number”);  
} else {  
    System.out.println(“odd number”);  
}
```

b) Rewrite the if-else as a switch statement

```
switch (num % 2) {  
    case 0:  
        System.out.println(“even number”);  
        break;  
    case 1:  
        System.out. println(“odd number”);  
        break;
```

}

- 4) Write statements that use Math.random() to generate random numbers for each of the following situations:

- a) Generate a random integer between 1 and 50
`int num = (int)(Math.random() * 50) + 1;`
- b) Generate a random integer between 20 and 100
- c) `int num = (int)(Math.random() * 81) + 20;`
- d) Generate a random double between 10 and 20, inclusive
`double num = Math.random() * 10 + 10;`

- 5) Identify the logic errors in the statements below, which should display a single appropriate message for any value of age:

```
if (age < 18) {  
    System.out.println("child");  
} else if (age > 18 && age < 65) {  
    System.out.println("adult");  
} else if (age > 65) {  
    System.out.println("senior");  
}
```

Age 18 and age 65 are not covered because the conditions use >18> and >65>
It should have used <= or >= statements

- 6) Given the following assignments, determine if each of the following expressions evaluates to true or false:

size = 100 weight = 50 value = 75

- a) size > 50 && weight == 50
true
- b) value < 100 && !(weight == 50)
false
- c) size >= 100 || value >= 100
false
- d) weight < 50 || size > 50
false
- e) !(value < 75)
true
- f) !(size > 100 && weight > 50 && value > 75)
true
- g) (value < 125 || weight < 76) && size == 100
false

- 8) Determine if each of the following are true or false. If false, explain why.

- a) The condition of an if statement must be a boolean expression
True, java's if statement is either true or false

- b) A nested if statement and an if-else statement are the same
False, a nested if statement is an if statement inside another if statement
- c) The expression in a switch statement must evaluate to a double
False, java switch does not use double
- d) Numbers generated by a computer program are actually pseudorandom
True, they are generated by algorithms
- e) The (double) cast is needed to generate a random integer
False, (int) is used to convert numbers into integers
- f) A compound Boolean expression can contain more than two Boolean expressions
True, e.g. `age > 18 && age < 65 && height < 170`
- g) In a logical And expression, both operands must be true for the expression to evaluate to true.
True, && requires both both sides to be true
- h) In logical expressions, && is evaluated before !.
False, ! takes priority
- i) The pow() method in the Math class is used for exponentiation
True, is means to the power of
- j) The statement `x = abs(-3);` will return the value 3.
False, the absolute value of -3 is 3